

1. Diket: larutan asam sulfat = 200 ml $M_r = 98 \text{ g/mol}$

Konsentrasi = 20%

Kerapatan = 1,2 g/ml

Dit: Kemolalan?

Jwb: massa asam sulfat = 20% · 200 ml = 40 g

$$\text{jumlah mol} = \frac{40 \text{ g}}{98 \text{ g/mol}} = 0,4082 \text{ mol}$$

$$\text{massa pelarut} = 200 \text{ ml} \cdot 1,2 \text{ g/ml} = 240 \text{ g} = 0,24 \text{ kg}$$

$$\text{kemolalan} = \frac{\text{jumlah mol}}{\text{massa pelarut}} = \frac{0,4082}{0,24 \text{ kg}} = 1,7012 \text{ mol/kg}$$

2. Diket: larutan asam sulfat = 200 ml

Konsentrasi = 20%

Kerapatan = 1,2 g/ml

$M_r = 98 \text{ g/mol}$

Dit: kemolaran

Jwb: massa asam sulfat = 20% · 200 ml = 40 g

$$\text{jumlah mol} = \frac{40 \text{ g}}{98 \text{ g/mol}} = 0,4082$$

$$\text{volume larutan} = 200 \text{ ml} = 0,2 \text{ L}$$

$$\text{molaritas} = \frac{0,4082}{0,2} = 2,041 \text{ M}$$

3. Diket: melarutkan 339 $\text{C}_2\text{H}_5\text{OH}$

100 ml larutan

kerapatan murni = 0,785 g/ml

Dit: konsentrasi $\text{C}_2\text{H}_5\text{OH}$ dinyatakan dengan % volume

$$\text{Jwb: } \frac{\text{massa}}{\text{volume}} = \frac{339}{\text{volume}} = 0,785 \text{ g/ml} \Rightarrow \text{volume} = \frac{339}{0,785 \text{ g/ml}} = 42,04 \text{ ml}$$

$$\left(\frac{\text{Volume alkohol}}{\text{Volume larutan}} \right) \times 100\%$$

$$= \frac{42,04 \text{ ml}}{100 \text{ ml}} \times 100\%$$

$$= 42,04 \text{ ml} = 42,04\% \text{ ml}$$

Imagination is never on the menu, you have to own the restaurant

No
Date

4. ~~Cairan~~ Diket: Cairan murni = 50 mm Hg pada 25°C

Dit: tekanan uap bila 6 mol dicampur dengan 4 mol suatu non elektrolit yg tidak menguap

Jawab: fraksi mol cairan murni = $\frac{6}{(6+4)} = \frac{6}{10} = 0,6$

Fraksi mol yang tidak menguap = $1 - 0,6 = 0,4$

Tekanan uap total campuran: $(0,6 \cdot 50) + (0,4 \cdot 0)$

= 30 mmHg